

DLA Draft

Developmental Learning Architecture: Learning History, Fast-Weight Traces, and Future Adaptation

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Abstract

A conventional neural learner is treated as a fixed function trained once by an external optimizer. This paper studies a more developmental question: does a learner’s prior learning history change how it will learn in the future, and which internal state carries this effect? We propose a Developmental Learning Architecture (DLA) that keeps standard MLP/Transformer backbones unchanged but augments each weight with per-parameter fast weights, plasticity, and sleep consolidation. In a 6.59M Chinese character GPT we find that different curriculum histories lead to different future adaptation on an unseen domain, and that this effect is causally tied to the fast-weight state w_{fast} : replacing a hard→easy learner’s fast weights with an easy→hard learner’s significantly degrades future adaptation ($n=12$, gain@40 difference -0.019 , bootstrap CI excludes zero, Cohen’s $d \approx -0.7$). Norm-matching and shuffling do not remove the effect, so it is not explained by weight magnitude or random parameter noise. The effect replicates on a second Shakespeare character-level backbone. We do not find evidence that more experience monotonically accelerates learning: multiple controlled longitudinal studies return null results. The conclusion is that learning

history leaves a persistent, partially localizable trace in transient learner state, with w_{fast} providing a causal component of history-dependent future adaptation.

1. Introduction

1.1 What problem exists

Modern neural networks are trained and then frozen. When they learn a second task, they either overwrite the first (catastrophic forgetting) or treat the two tasks as independent (Kirkpatrick et al., 2017; Zenke et al., 2017; De Lange et al., 2021). More fundamentally, standard training does not allow the model itself to become a different kind of learner: the learning algorithm is fixed outside the model, whether it is an external optimizer or a meta-learned rule (Andrychowicz et al., 2016; Miconi et al., 2019; Beaulieu et al., 2020).

We ask a problem that sits before “continual learning” and “meta-learning” — yet is closely tied to a long-standing empirical question about task and curriculum ordering (Poirier & Silver, 2005; Bell & Lawrence, 2022; Li & Hiratani, 2025):

Can what a model has experienced change how it will learn in the future, and which part of the model carries that developmental change?

1.2 Why it matters

If learning history only changes stored knowledge, the learner remains a static function with a growing database. If learning history also changes the learner state, then a small model could, in principle, become a better learner through experience—without changing its architecture. This developmental framing has a deep biological precedent: complementary learning systems (CLS) theory explains memory/learning trade-offs by the interaction of a fast hippocampal system and a slow neocortical system, consolidated over time (McClelland, McNaughton & O'Reilly, 1995; Kumaran, Hassabis & McClelland, 2016). Recent empirical results further show that plasticity itself is a fragile resource that ordinary training can consume (Dohare et al., 2024; Lyle, Rowland & Dabney, 2022; Nikishin et al., 2022), so “what kind of learner a model is” is not

fixed by its architecture alone. This is the key motivation for “growing models” and for understanding whether development is a meaningful object in machine learning.

The practical significance is twofold. First, it tells us which parts of a model to preserve or transfer between learning phases. Second, it gives a falsifiable framework for claims such as “more experience makes models learn faster” — a claim we test and do not find support for.

2. How existing work addresses the problem

- **Catastrophic forgetting methods** (regularization, replay, EWC, pseudorehearsal) address memory, not the change of the learner itself (Kirkpatrick et al., 2017; Zenke et al., 2017; Robins, 1995; De Lange et al., 2021).
- **Fast-weight models** add a temporary memory but usually keep the learning rule external. The idea goes back to fast-weight memories (Hinton & Plaut, 1987; Ba et al., 2016) and has seen a modern revival: linearised attention is formally equivalent to fast-weight programming (Schlag, Irie & Schmidhuber, 2021), in-context learning can be understood as implicit gradient descent / fast-weight updates inside a Transformer (von Oswald et al., 2023), and recent sequence models introduce trainable “neural memory” modules with fast updates alongside slow base weights (Behrouz, Zhong & Mirrokni, 2025). Fast/slow dual-module designs also appear in RLHF pipelines (“fast-slow chasing” in online DPO; Qi et al., 2024). In most of these, the fast store is either a memory mechanism or an engineering device; DLA instead treats fast/slow as a *developmental state* whose history we causally dissect.
- **Complementary learning systems and consolidation.** CLS theory attributes the complementary benefits of fast, instance-based and slow, generalising stores to their interaction and to offline consolidation (McClelland et al., 1995; Kumaran et al., 2016). Machine-learning instantiations usually realise this as experience replay; DLA instantiates the fast/slow split as an internal, per-parameter weight state with sleep-boundary consolidation and no external replay buffer.
- **Meta-learning** optimizes a learning rule over many episodes; it asks whether a rule can be learned, but rarely whether one individual’s rule develops within a single lifetime (Andrychowicz et al., 2016; Beaulieu et al., 2020).
- **Learned plasticity** (differentiable plasticity, ANML) shows that plasticity can be meta-learned, but does not separate *body state*

from *learning-rule parameters* as causal carriers (Miconi et al., 2019; Beaulieu et al., 2020). Complementary findings that plasticity degrades over long training—capacity loss and primacy bias—motivate making plasticity itself a state to maintain (Dohare et al., 2024; Lyle et al., 2022; Nikishin et al., 2022).

- **Task and curriculum ordering.** The order in which tasks are learned measurably changes continual-learning outcomes and the amount of forgetting (Bell & Lawrence, 2022; Li & Hiratani, 2025), affects the consolidation of task knowledge (Poirier & Silver, 2005), and is the core object of curriculum learning (Wang, Chen & Zhu, 2022). This literature mostly measures order effects on *stored knowledge* about seen tasks. We ask a complementary question: does order change the *learner state* itself, such that future adaptation to a *never-seen* task changes—and which state component carries that effect?
 - **We therefore design DLA** to ask a mechanistic question that is usually skipped: given the same task sequence, which state component carries history-dependent future adaptation?
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3. Our solution: DLA

3.1 Design principle

We do not modify the Transformer skeleton. Instead, each weight matrix is accompanied by:

- W_{slow} : long-term consolidated knowledge
- W_{fast} : a fast learning trace
- P : per-parameter plasticity
- Q : sleep eligibility trace

Effective weights:

$$W_{\text{eff}} = W_{\text{slow}} + \text{softplus}(P) * W_{\text{fast}}$$

Structurally, this additive form resembles parameter-efficient transfer methods that freeze a base network and attach small trainable additive weights—adapters (Houlsby et al., 2019), LoRA (Hu et al., 2021), and the unified additive-PEFT view (He et al., 2022). DLA differs in three ways that matter for development: (i) the additive pathway is a *within-lifetime developmental state* updated online by the learner’s own wake/sleep dynamics, not a separately fine-tuned module; (ii) it is shared across tasks and persists (subject to decay and consolidation) rather than being re-initialised per task; and (iii) its contribution is gated per parameter by

a learned plasticity $\text{softplus}(P)$. The fast/slow decomposition itself follows complementary learning systems theory (McClelland et al., 1995; Kumaran et al., 2016).

3.2 Wake and sleep

During learning, W_{fast} is updated with Adam-style moments, gated by $\text{softplus}(P)$. At task boundaries (sleep), part of W_{fast} is consolidated into W_{slow} through Q , and W_{fast} is decayed. This is the fast/slow separation: it follows the CLS prescription that fast, recent traces be gradually transferred into a slow, generalising store (McClelland et al., 1995; Kumaran et al., 2016), but is implemented as internal weight state with no external replay buffer. Fast/slow weight co-design has engineering precedent in optimisation and RLHF (Qi et al., 2024) and in neural-memory architectures (Behrouz et al., 2025); here it is studied as a developmental mechanism whose causal carrier we test. **Algorithm 1: DLA wake-sleep cycle**

```
1: for task  $t$  in curriculum do
2:   for step in  $1..T$  do
3:     compute  $g = dL/dW_{\text{eff}}$ 
4:     update  $W_{\text{fast}}$  with Adam( $g$ ), gated by  $\text{softplus}(P)$ 
5:     update  $P$  from progress/relevance
6:   end for
7:   consolidate:  $W_{\text{slow}} += \beta * Q + \gamma * W_{\text{fast}}$ 
8:   decay:  $W_{\text{fast}} *= c$ ,  $Q *= d$ 
9: end for
```

3.3 Experimental design to isolate development

To test whether history changes the future learner, we:

1. Build two histories with the same three domains but different order:
 - easy→hard (EH)
 - hard→easy (HE)
 2. Probe each individual on a never-seen domain D .
 3. Perform body/component cross-injection to see which state component transfers the history effect.
 4. Run norm-matched, shuffled, and module-localized controls to rule out magnitude/randomness.
 5. Compare with standard baselines (AdamW, replay, EWC).
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4. Experimental setup

Primary backbone: 6.59M Chinese character GPT (6 layers, 8 heads, 256 dim, vocab 7280), pretrained on Chinese Wikipedia. Curriculum domains: Wikipedia, SFT–style QA, science Wikipedia; unseen domain D is a science slice. Second setting: ~10.65M Shakespeare character GPT (vocab 65). Metrics: gain@40, LE_D, T80, paired bootstrap CI, sign test.

Table 1: Common experimental hyperparameters.

Hyperparameter	Value
Block size	128
Batch size	32
Steps per task	40
AdamW learning rate (baselines/DLA base)	1e−4
EWC lambda	1e3
Replay buffer capacity	48 batches
Seeds (primary setting)	12
Seeds (baselines)	5
Seeds (second setting)	2

5. Results: how well does it work

5.1 Fast/slow separation protects consolidated memory

DLA matches tuned AdamW on new–domain adaptation (gain +14.0% vs +13.0%) while slow–memory forgetting is ≈ 0 (−0.4%). This shows the mechanism does not sacrifice memory for plasticity.

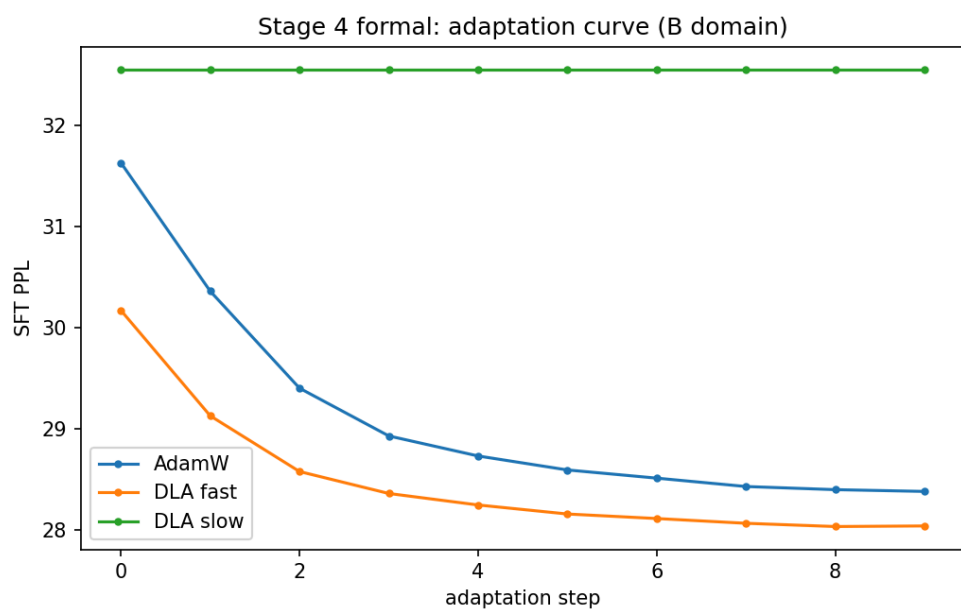
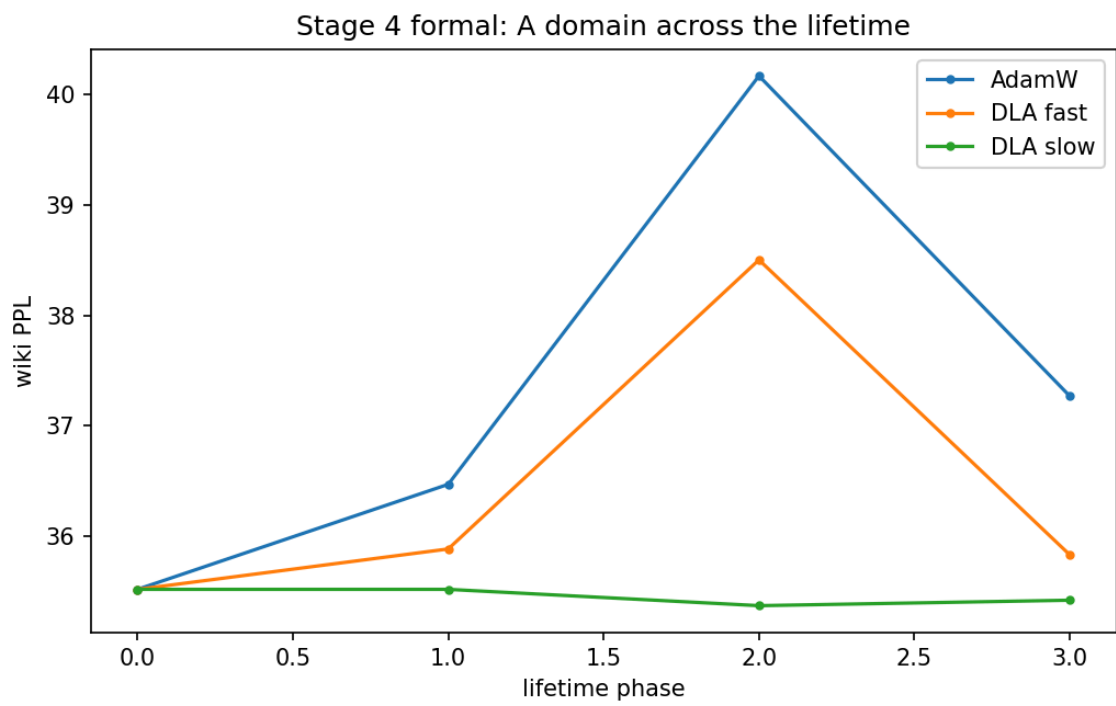


Figure 1b: Adaptation curve on the new domain.

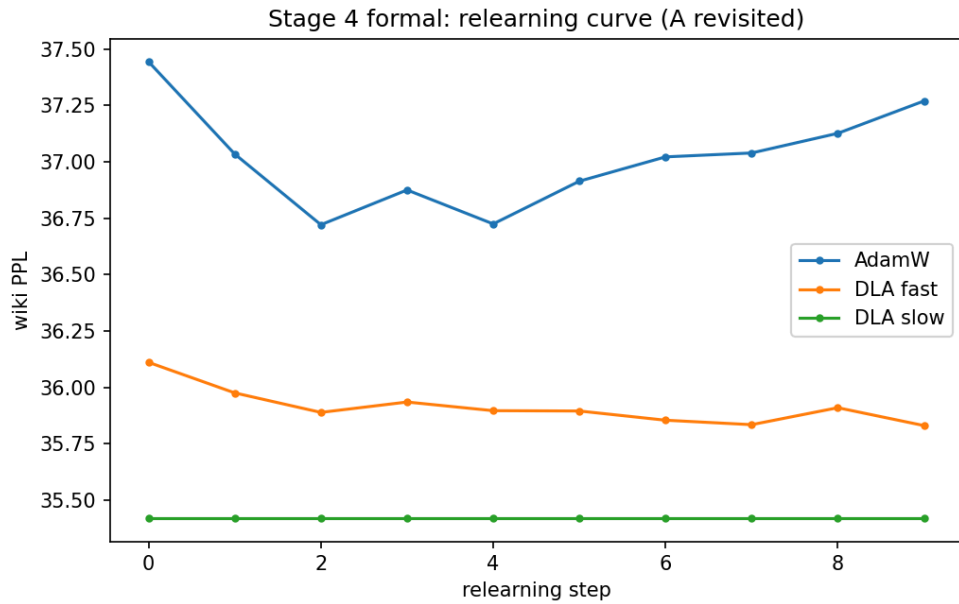


Figure 1c: Relearning curve.

5.2 Learning history changes future adaptation

Across histories, HE learners adapt better to unseen D than EH learners. That later adaptation depends on the *order* of earlier tasks mirrors continual-learning and curriculum results measured on seen tasks (Bell & Lawrence, 2022; Li & Hiratani, 2025; Poirier & Silver, 2005). The distinctive feature here is that D was never seen in either history, so the order effect must act through the learner’s internal state rather than through stored content about D. This is not unique to DLA; but DLA is most robust after a difficult history.

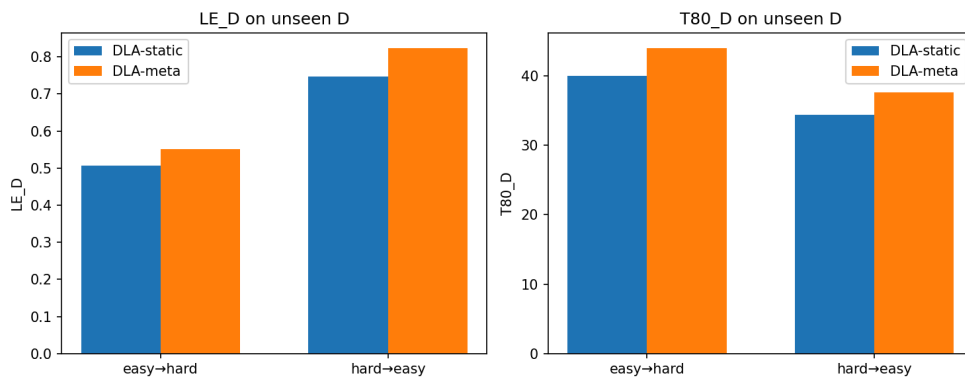


Figure 2: History effect on unseen D (2x2 development experiment).

5.3 The effect is not carried by the learning-rule parameters

ϕ

Body $\times\phi$ cross-injection shows that swapping ϕ between histories changes future adaptation much less than swapping body state. ϕ is not a stable/dominant carrier in this setting.

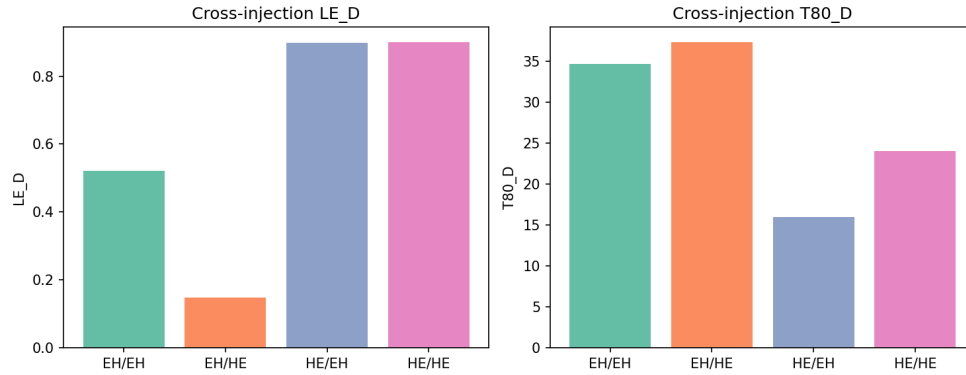


Figure 3: Body $\times$ ϕ causal dissection.

5.4 W_fast is a causal component

Core result (n=12):

condition	mean gain@40
HE/HE	+0.014
HE + EH W_fast	-0.005

Paired HE $\rightarrow$ raw EH: mean -0.019 , 95% CI $[-0.033, -0.005]$, Cohen's $d \approx -0.71$, sign $p=0.019$ (10/12 negative). Initial PPL differences are small; the effect appears in the adaptation trajectory.

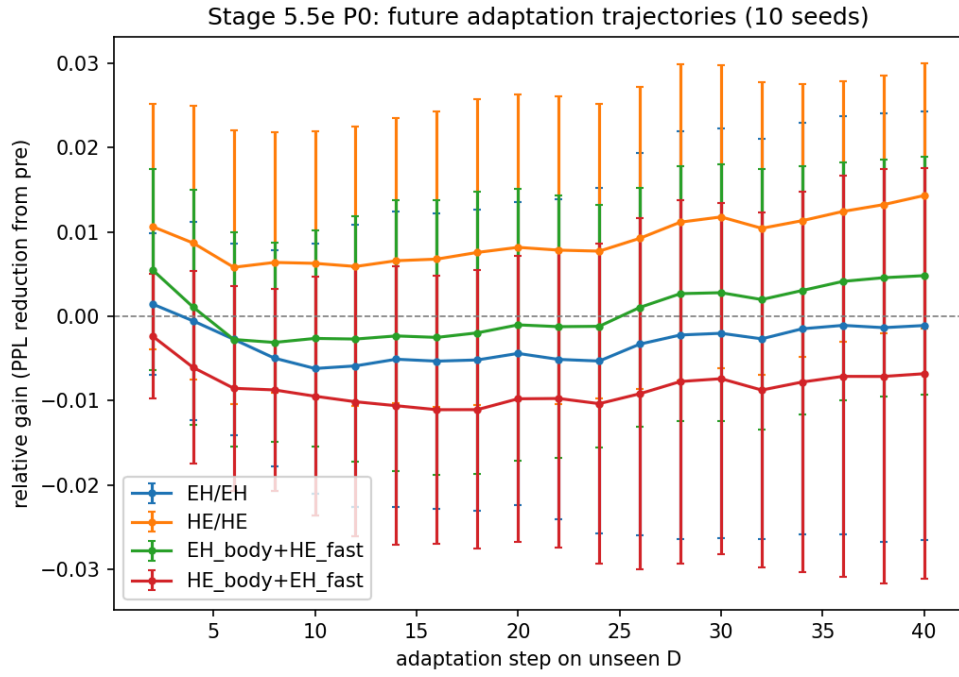


Figure 4: P0 future-adaptation trajectories.

5.5 Controls: magnitude, randomness, module

control	mean Δ gain@40	95% CI	d
raw EH W_fast	-0.019	[-0.034, -0.005]	-0.71
norm-matched EH W_fast	-0.022	[-0.034, -0.009]	-0.90
shuffled EH W_fast	-0.017	[-0.023, -0.009]	-1.25
EH embedding W_fast	-0.006	[-0.009, -0.002]	-0.87
EH attention W_fast	-0.012	[-0.018, -0.006]	-1.06
EH MLP W_fast	-0.017	[-0.025, -0.008]	-1.05

Norm-matching does not rescue the effect; shuffling also does not remove it. Module localization is suggestive of MLP and attention, with embedding weaker.

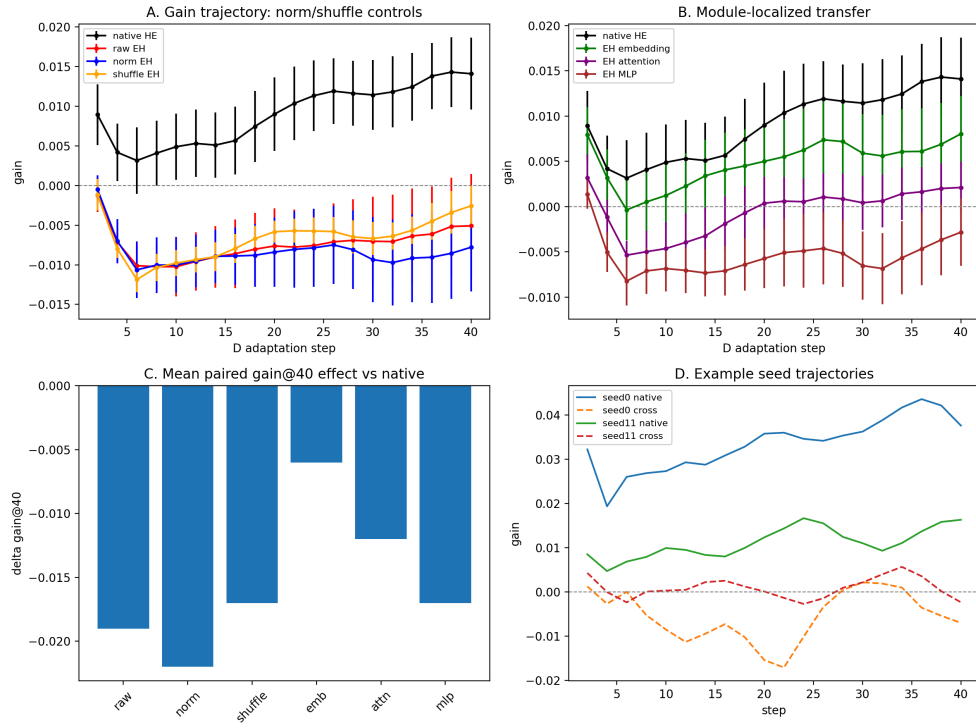


Figure 5: W_{fast} cross-injection and full future-adaptation trajectories. A: norm/shuffle controls. B: module-localized transfer. C: paired gain@40 effects. D: example trajectories.

5.6 Fair baselines

method	EH	HE	HE-EH
AdamW	0.339	0.675	+0.336
AdamW+replay	0.375	0.961	+0.586
EWC	0.292	0.895	+0.603
DLA	0.517	0.709	+0.192

DLA is the most robust after an easy $\rightarrow$ hard history; standard learners show larger HE performance but larger EH degradation.

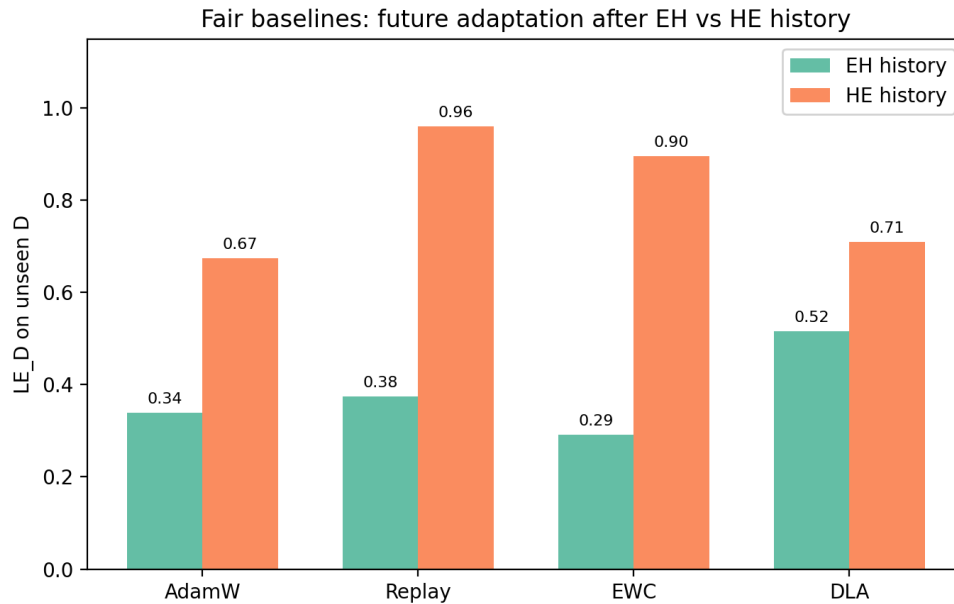


Figure 6: Fair baselines — future adaptation after EH vs HE history.

5.7 Second-setting replication

Shakespeare char GPT, 2 seeds: HE/HE gain@40 +0.052/+0.042; HE+EH fast +0.004/−0.001. The destructive effect replicates in direction.

5.8 Negative result: more experience does not necessarily make learning faster

Stage 5 (difficulty-normalized): flat LE. Stage 6 (matched difficulty): $p=0.17$. Stage 7 (cross-domain, 20 seeds): $p=0.82$. Stage 8 (physics near-transfer, 20 seeds): $d=-0.17$. We therefore do not claim “more experience → faster learning”. This null is consistent with a growing plasticity literature: more training does not monotonically make a network more learnable, early experience can dominate later learning (primacy bias; Nikishin et al., 2022), and both capacity and plasticity can be consumed by ordinary updates (Lyle et al., 2022; Dohare et al., 2024).

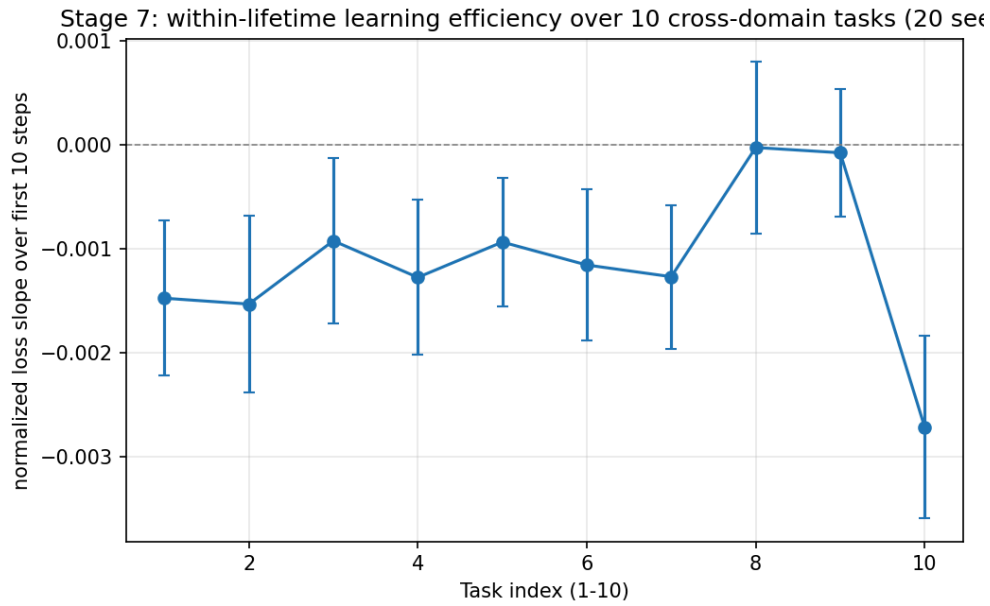


Figure 7a: Negative result — cross-domain longitudinal (Stage 7, 20 seeds).

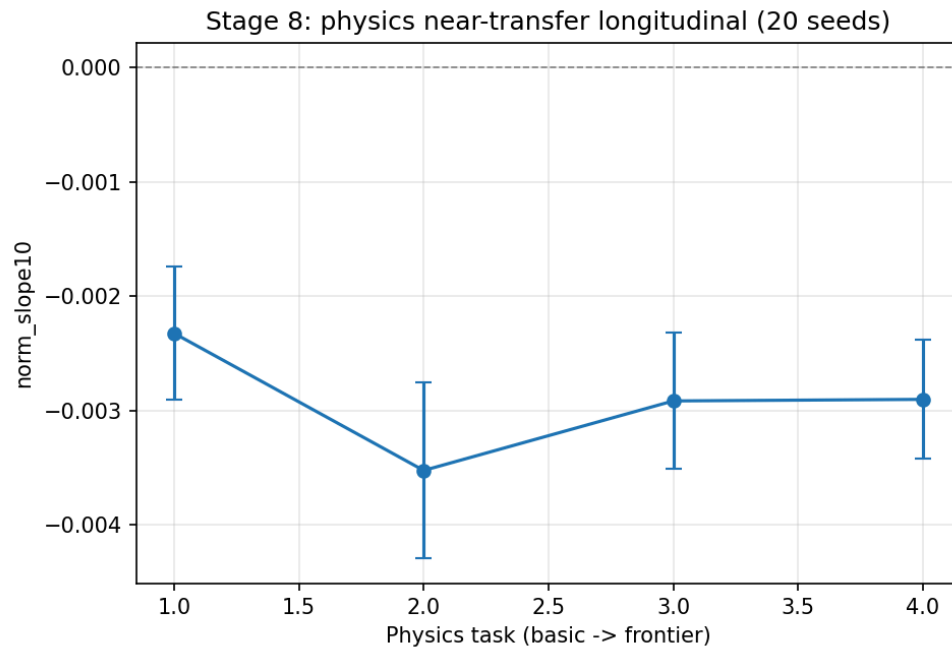


Figure 7b: Negative result — physics near-transfer (Stage 8, 20 seeds).

6. Limitations

- Second-setting $n=2$; baselines $n=5$.
- Single small backbones; no large-scale validation.
- Sleep consolidation writes only a small amount into w_{slow} ; fast/slow separation is mostly isolation.

- No pre-registration; we report $n=10 \rightarrow n=12$ transparently.
 - We cannot yet claim W_{fast} is the only carrier or that module localization is definitive.
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7. Conclusion and significance

The learner is not fully characterized by what it currently knows. Learning history leaves a persistent, partially localizable trace in transient learner state, with w_{fast} providing a causal component of history-dependent future adaptation. This result gives a concrete mechanism for “development” in neural networks and clarifies a frequent but unsupported claim: developmental change does not equal universally faster learning.

Reproducibility

Code: <https://github.com/JayCRL/DLA> Experiment details: docs/experiments.md, paper/overnight_report.md, paper/validation_audit.md Scripts: stage4_formal.py, stage5e_wfast_p0.py, validation_p0_controls.py, validation_baselines.py, validation_second_setting.py, stage6_longitudinal.py, stage7_cross_domain.py, stage8_physics.py.

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